

RSA example

- Choose 2 primes
- $p = 7$
- $q = 11$
- Compute $n = pq$
- Get Φ of $n = \Phi_{22}(p-1)(q-1)$
- Choose public exponent e
 e must be coprime of $\Phi(n)$
- Compute private exponent d
 $d = e^{-1} \bmod \Phi(n)$

Public key
 (e, n)

Private is
 (d, n)

- To encrypt $C = m^e \bmod n$
- To decrypt $C^d = m^{ed} \bmod n$